

SDITO-LAB

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Roll No : CSE/23/080

Exam Roll No :11600223051

Assignment NO : 09

Problem Statement : Deploy a Project from GitHub to AWS EC2 Instance

Aim

To launch an EC2 instance, connect it via SSH, install required software, clone a project from GitHub, and deploy it so it is accessible via Public IPv4 address.

Requirements

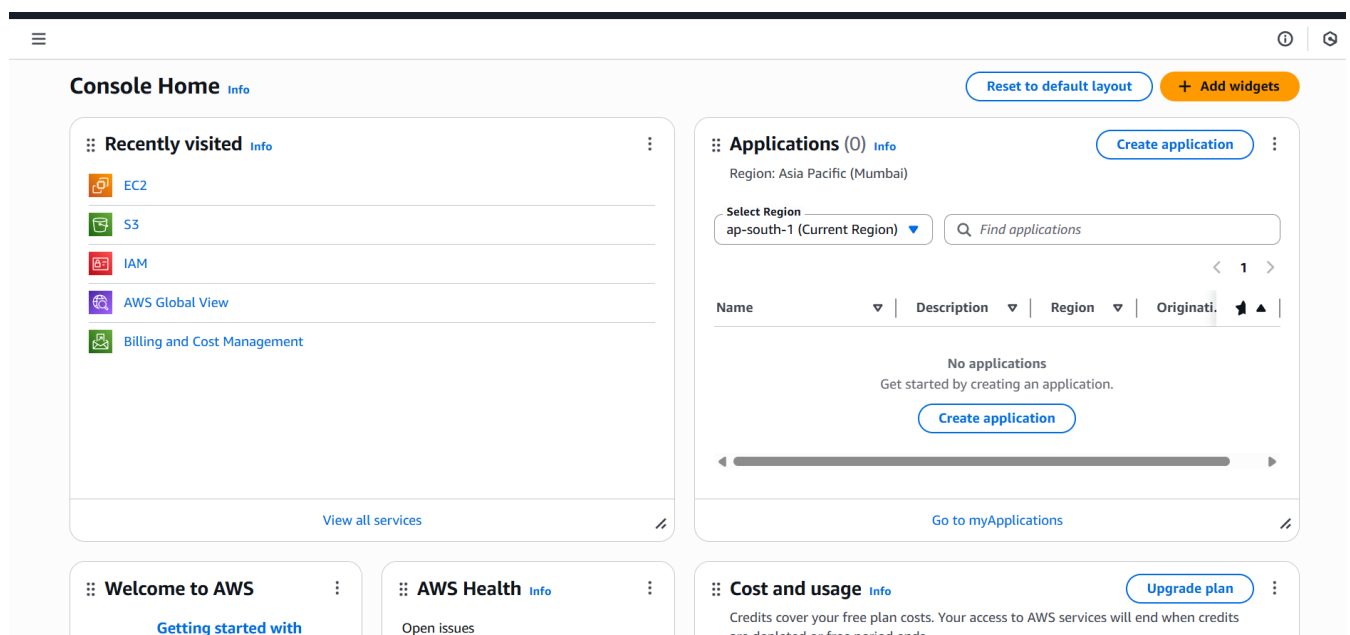
- AWS Account
- EC2 Ubuntu Instance
- Key Pair (.pem)
- GitHub Repository
- Node.js Project
- Security Group with required ports open

Solution:

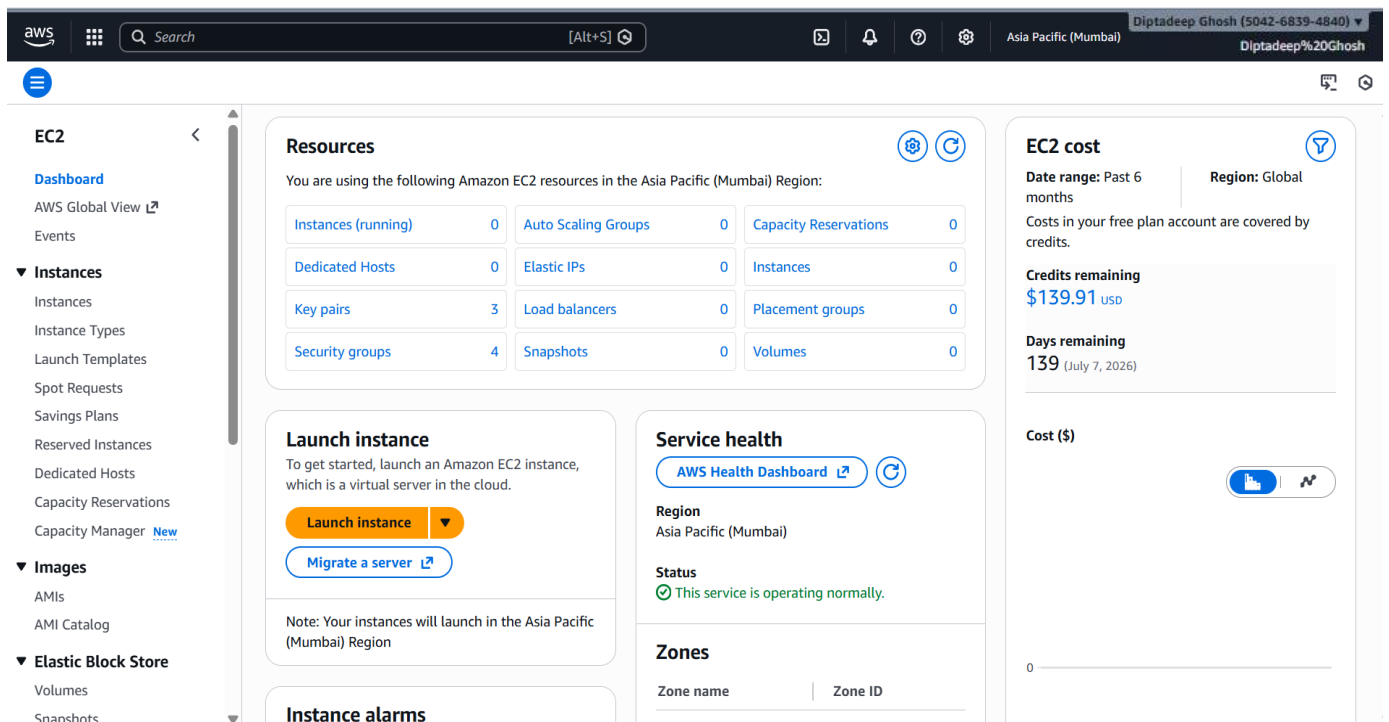
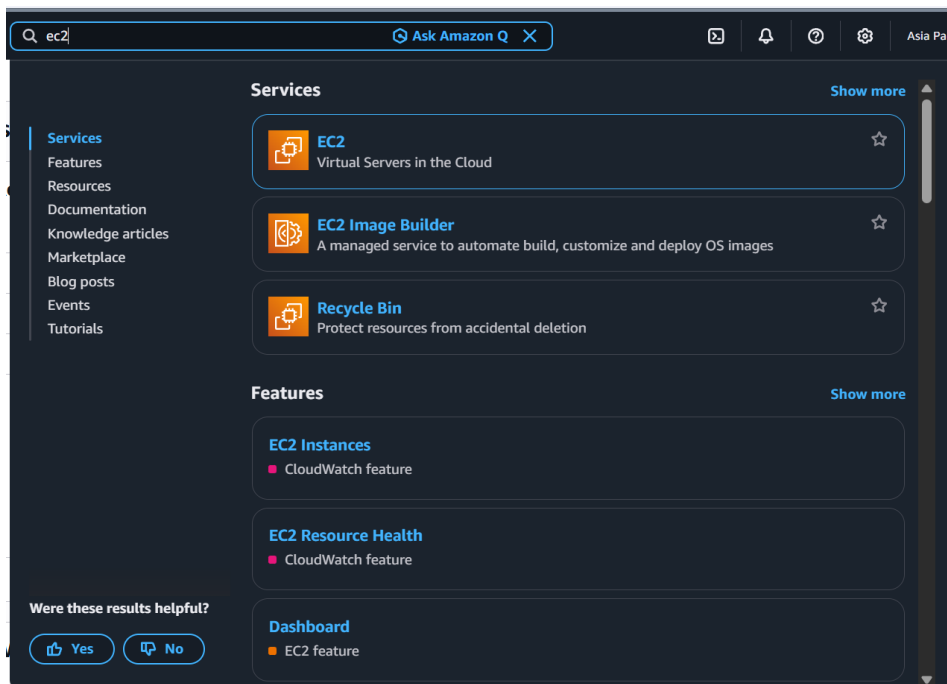
PHASE 1 – Launch EC2 Instance

Step 1: Launch an EC2 Instance

1. Login to AWS Management Console.



2. Search for EC2 and open the service.



3. Click Launch instance.

Instance Configuration

4. Enter:

- Name: WebServer-ASSG9

5. Select:

- AMI: Ubuntu (Free Tier Eligible)

aws [Search] [Alt+S] Asia Pacific (Mumbai) Diptadeep Ghosh (5042-6839-4840) Diptadeep%20Ghosh

EC2 > Instances > Launch an instance

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

WebServer-ASSG9 [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

Summary

Number of instances [Info](#)

1

[Software Image \(AMI\)](#)
Amazon Linux 2023 AMI 2023.10...[read more](#)
ami-0317b0f0a0144b137

[Virtual server type \(instance type\)](#)
t3.micro

[Firewall \(security group\)](#)
New security group

[Storage \(volumes\)](#)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

6. Choose:

- Instance type: t3.micro

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EC2 > Instances > Launch an instance

Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro [Free tier eligible](#)

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0112 USD per Hour

On-Demand SUSE base pricing: 0.0112 USD per Hour

On-Demand Windows base pricing: 0.0204 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour

On-Demand RHEL base pricing: 0.04 USD per Hour

[Additional costs apply for AMIs with pre-installed software](#)

☐ All generations [Compare instance types](#)

Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Select [Create new key pair](#)

Network settings [Info](#)

Network [Info](#)

vpc-035f96df3c676ba1d [Edit](#)

Summary

Number of instances [Info](#)

1

[Software Image \(AMI\)](#)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-019715e0d74f695be

[Virtual server type \(instance type\)](#)
t3.micro

[Firewall \(security group\)](#)
New security group

[Storage \(volumes\)](#)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

7. Click Create new key pair:

- Name: WEBKEY
- Type: RSA
- Format: .pem

aws [Search] [Alt+S] Asia Pacific (Mumbai) Diptadeep Ghosh (5042-6839-4840) Diptadeep%20Ghosh

EC2 > Instances > Launch an instance

Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro [Free tier eligible](#)

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0112 USD per Hour

On-Demand SUSE base pricing: 0.0112 USD per Hour

On-Demand Windows base pricing: 0.0204 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour

On-Demand RHEL base pricing: 0.04 USD per Hour

[Additional costs apply for AMIs with pre-installed software](#)

Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance.

Key pair name - *required*

Select

Network settings [Info](#)

Network [Info](#)

vpc-035f96df3c676ba1d

Create key pair

Key pair name

Key pairs allow you to connect to your instance securely.

WEBKEY

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

[When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. \[Learn more\]\(#\)](#)

[Cancel](#) [Create key pair](#)

8. Download and save the key file securely.
9. Click on **Create security group** and check the boxes of **Allow HTTPS traffic from the internet** and **Allow HTTP traffic from the internet** and click on **Launch instance** button.

Network settings [Info](#) [Edit](#)

Network [Info](#)
vpc-035f96df3c676ba1d

Subnet [Info](#)
No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)
Enable

Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-4' with the following rules:

- ☒ Allow SSH traffic from
Helps you connect to your instance
Anywhere
0.0.0.0/0
- ☒ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server
- ☒ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only. ✕

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-019715e0d74f695be

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

10. The Instance will be successfully created , return to the **Instance** page.

Success
Successfully initiated launch of instance (i-00ee2b4655a95b9a5)

► **Launch log**

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
[Create billing alerts](#)

Connect to your instance
Once your instance is running, log into it from your local computer.
[Connect to instance](#)
[Learn more](#)

Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
[Connect an RDS database](#)
[Create a new RDS database](#)
[Learn more](#)

Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
[Create EBS snapshot policy](#)

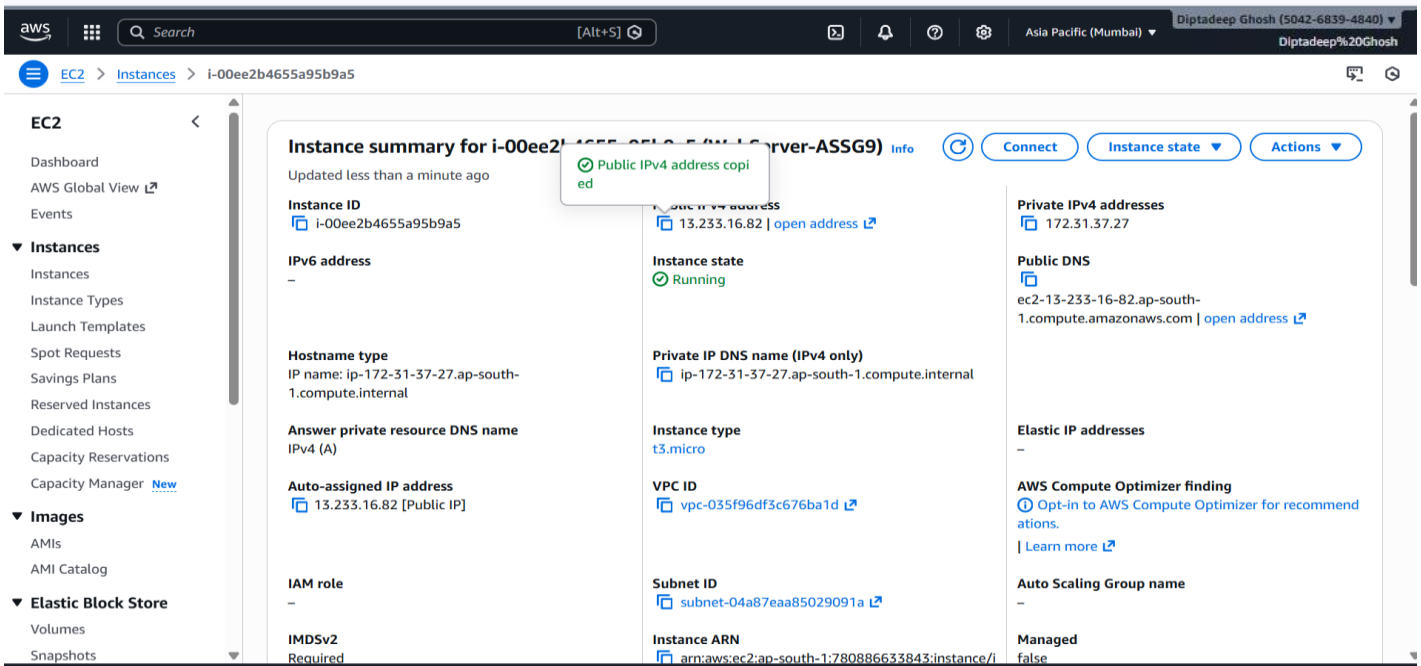
Manage detailed monitoring

Create Load Balancer

Create AWS budget

Manage CloudWatch alarms

11. The Instance will be seen running, click on the newly created instance.



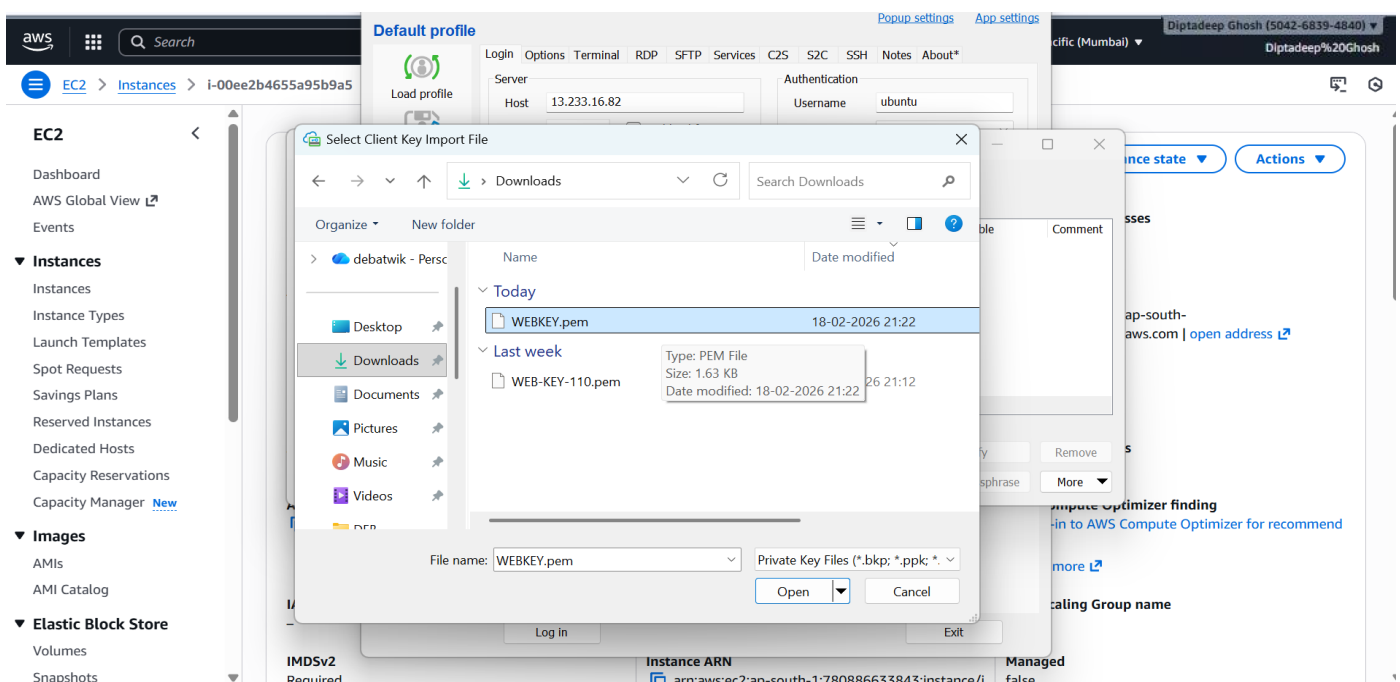
Step 2: Copy Public IPv4 Address

After the EC2 instance is launched and running, copy the Public IPv4 address from the Instance details page.

PHASE 2 – Connect to EC2 Instance

Step 3: Open Bitwise SSH Client and Configure Connection

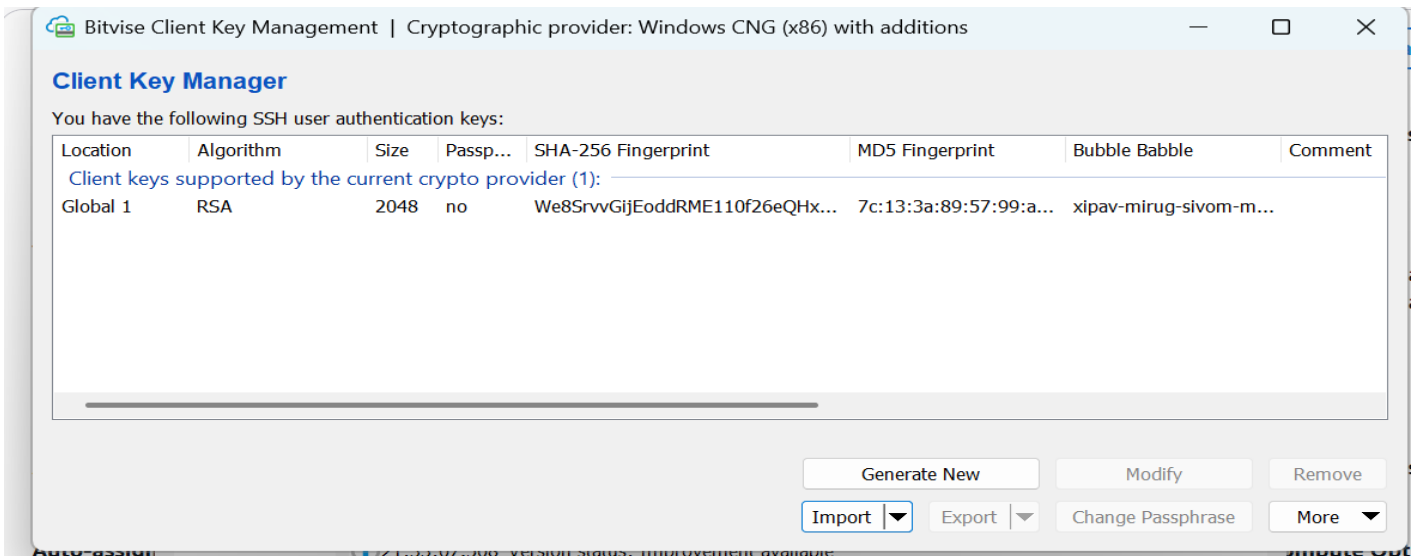
1. Open Bitwise SSH Client on the computer.
2. Paste the copied Public IPv4 address into the Host field.
3. Enter ubuntu in the Username field.
4. Set Initial method to publickey.
5. Click on Client key manager.



Step 4: Import Private Key

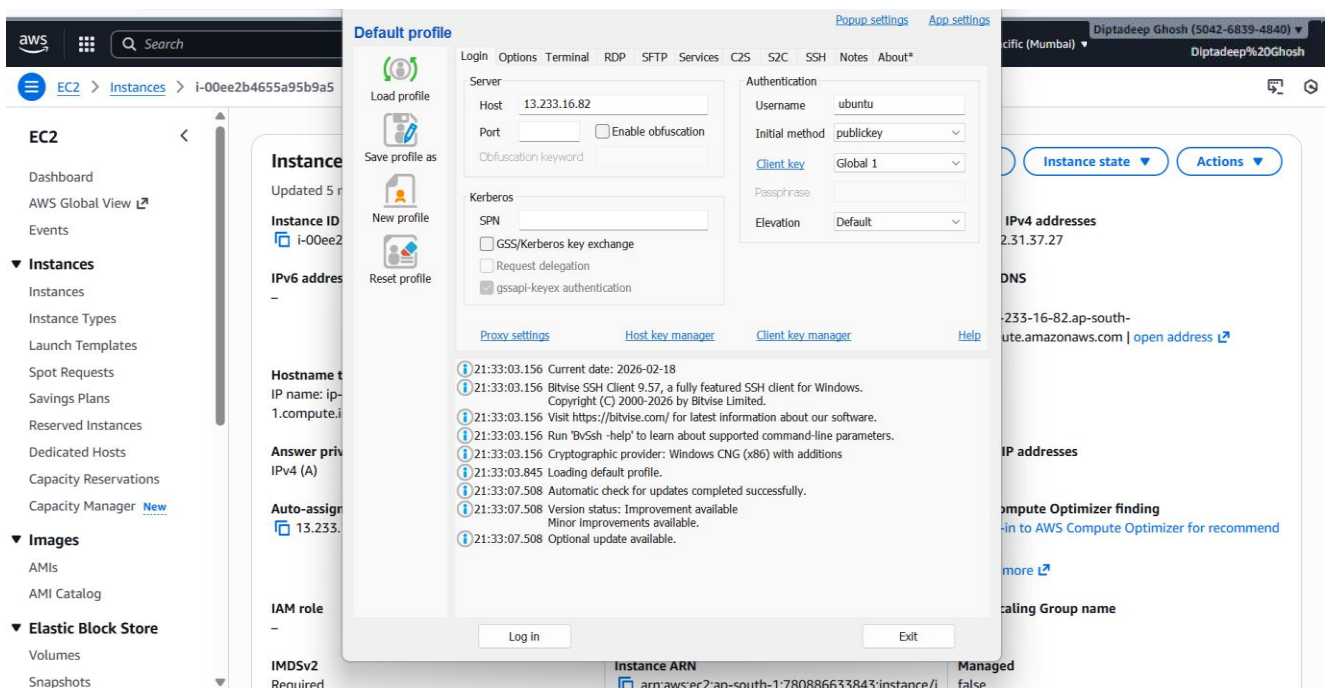
1. In Client Key Manager, click Import.
2. Select the downloaded .pem key file from the Desktop.

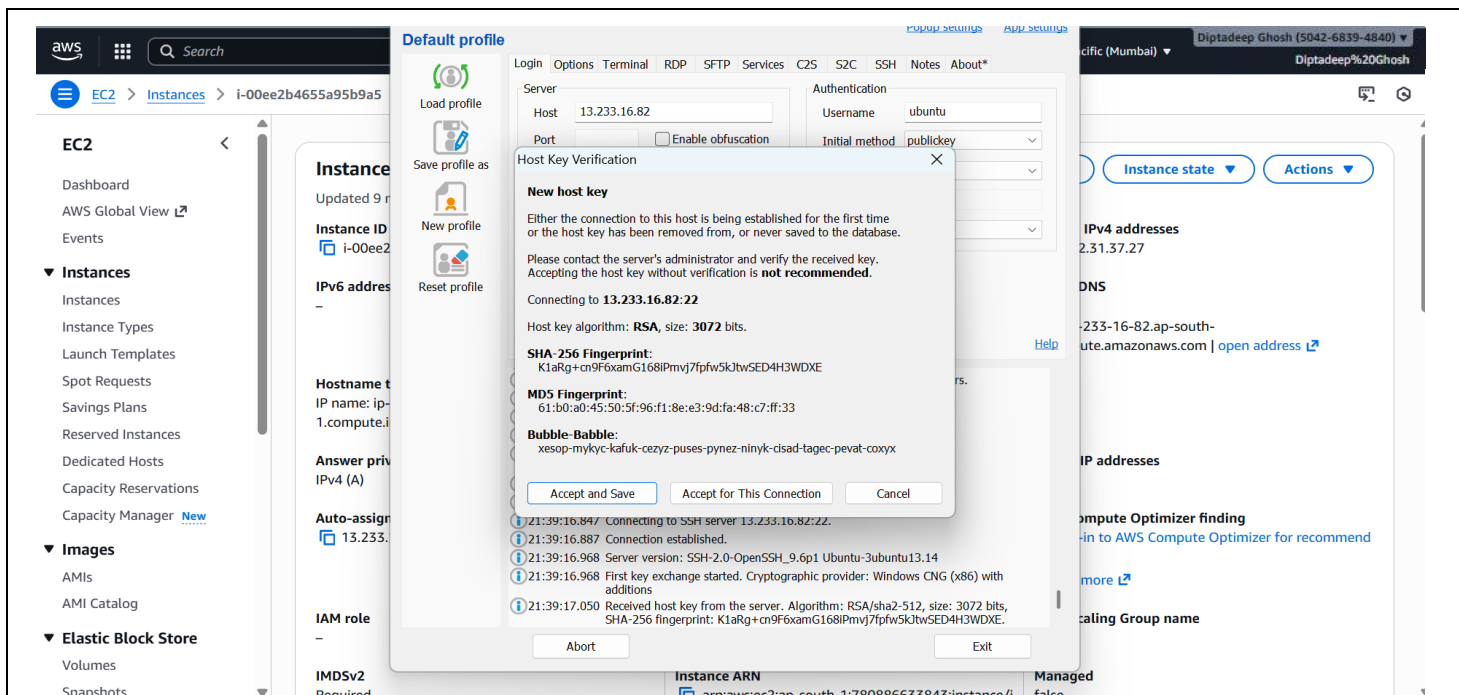
3. Save the key.



Step 5: Login to the Instance

1. Ensure Client key is set to Global 1.
2. Click on the Log in button.





PHASE 3 – Update Linux System and Install Required Software

Step 6: Open Terminal Console

After successful login:

- Click on New Terminal Console from the Bitvise side panel.

```
ubuntu@13.233.16.82:22 - Bitvise xterm - ubuntu@ip-172-31-37-27: ~
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro

System information as of Wed Feb 18 16:10:17 UTC 2026

System load: 0.0      Temperature: -273.1 C
Usage of /: 25.9% of 6.71GB   Processes: 110
Memory usage: 24%      Users logged in: 0
Swap usage: 0%          IPv4 address for ens5: 172.31.37.27

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-37-27:~$
```

Step 7: Update Ubuntu System

Run the following command to update and upgrade the system:

sudo apt-get update && sudo apt-get upgrade -y

```
ubuntu@ip-172-31-37-27:~$ sudo apt-get update && sudo apt-get upgrade -y
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:9 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:11 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:12 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
]
```

```

Running kernel seems to be up-to-date.

Restarting services...
systemctl restart acpid.service chrony.service cron.service irqbalance.service packagekit.service polkit.service ssh.service udisks2.service

Service restarts being deferred:
systemctl restart ModemManager.service
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@ttyS0.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
ubuntu @ session #2: sshd[1058,1172]
ubuntu @ user manager service: systemd[1067]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
N: Some packages may have been kept back due to phasing.
ubuntu@ip-172-31-37-27:~$

```

Step 8: Install Nginx Web Server

Install Nginx using the command:

sudo apt-get install nginx -y

```

ubuntu@ip-172-31-37-27:~$ sudo apt-get install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 7 not upgraded.
Need to get 565 kB of archives.
After this operation, 1596 kB of additional disk space will be used.
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.6 [43.5 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.6 [521 kB]
Fetched 565 kB in 0s (22.4 MB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 71753 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.6_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.6) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.6_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.6) ...
Setting up nginx-common (1.24.0-2ubuntu7.6) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.

```

Step 9: Install Node.js (LTS Version)

To install the latest supported LTS version of Node.js on Ubuntu, the NodeSource repository is configured and Node.js is installed using the following commands:

Add NodeSource Repository (Node.js 20 LTS)

curl -SL https://deb.nodesource.com/setup_20.x | sudo -E bash -


```

ubuntu@ip-172-31-37-27:~$ curl -SL https://deb.nodesource.com/setup_20.X | sudo -E bash -
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100 3907 100 3907    0     0 17373    0 --:--:-- --:--:-- --:--:-- 17441
2026-02-18 16:28:44 - Installing pre-requisites
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
7 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ca-certificates is already the newest version (20240203).
ca-certificates set to manually installed.
curl is already the newest version (8.5.0-2ubuntu10.6).
curl set to manually installed.
gnupg is already the newest version (2.4.4-2ubuntu17.4).
gnupg set to manually installed.
The following NEW packages will be installed:
  apt-transport-https
0 upgraded, 1 newly installed, 0 to remove and 7 not upgraded.
Need to get 3970 B of archives.
After this operation, 36.9 kB of additional disk space will be used.
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 apt-transport-ht
ps all 2.8.3 [3970 B]
Fetched 3970 B in 0s (282 kB/s)
Selecting previously unselected package apt-transport-https.
(Reading database ... 71801 files and directories currently installed.)
Preparing to unpack .../apt-transport-https 2.8.3 all.deb ...

```

Install Node.js

sudo apt-get install -y nodejs

```

ubuntu@ip-172-31-37-27:~$ sudo apt-get install -y nodejs
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  nodejs
0 upgraded, 1 newly installed, 0 to remove and 7 not upgraded.
Need to get 32.0 MB of archives.
After this operation, 197 MB of additional disk space will be used.
Get:1 https://deb.nodesource.com/node_20.x nodistro/main amd64 nodejs amd64 20.20.0-1nodesource1 [32
.0 MB]
Fetched 32.0 MB in 1s (63.9 MB/s)
Selecting previously unselected package nodejs.
(Reading database ... 71805 files and directories currently installed.)
Preparing to unpack .../nodejs_20.20.0-1nodesource1_amd64.deb ...
Unpacking nodejs (20.20.0-1nodesource1) ...
Setting up nodejs (20.20.0-1nodesource1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@ttyS0.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

```

Verify Installation

```
node -v
```

```
npm -v
```

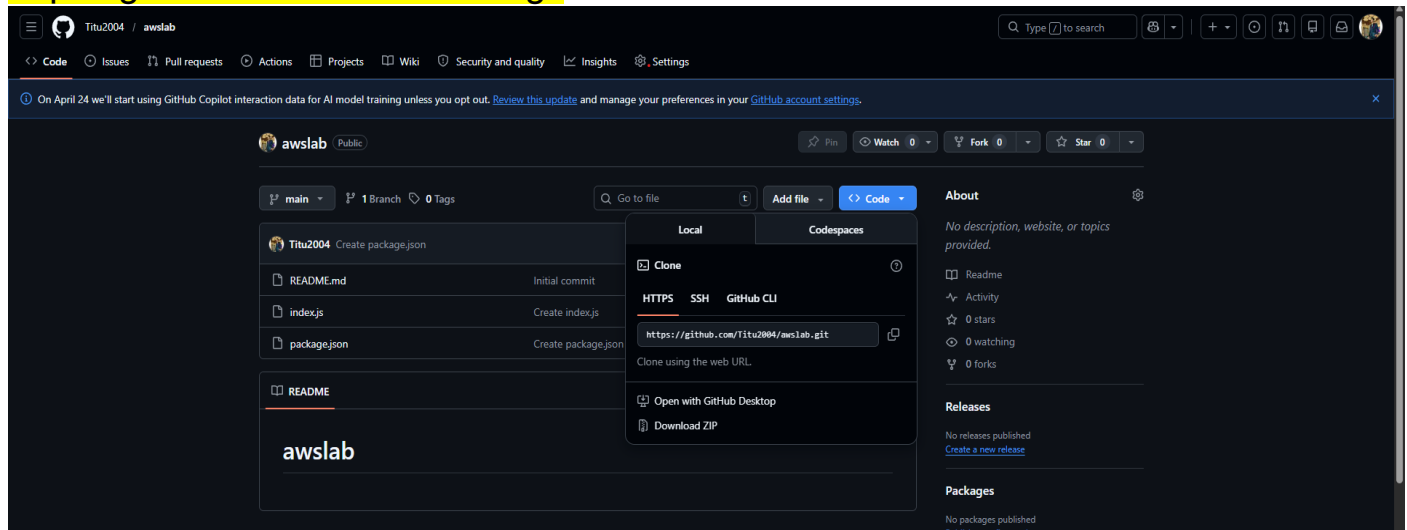
```
ubuntu@ip-172-31-37-27:~$ node -v
v20.20.0
ubuntu@ip-172-31-37-27:~$ npm -v
10.8.2
```

PHASE 4 – Clone GitHub Repository

Step 1: Copy Repository URL from GitHub

Go to GitHub → Open repository → Click Code → Copy HTTPS URL.

<https://github.com/titu2004/aws.git>



Step 2: Clone Repository in EC2

`git clone https://github.com/titu2004/aws.git` Move inside folder:

```
cd aws
```

```
ls
```

You should see:

index.js

package.json

```
ubuntu@ip-172-31-37-27:~$ git clone https://github.com/debatwiksantra/aws.git
Cloning into 'aws'...
remote: Enumerating objects: 15, done.
remote: Counting objects: 100% (15/15), done.
remote: Compressing objects: 100% (14/14), done.
remote: Total 15 (delta 5), reused 4 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (15/15), 4.15 KiB | 1.38 MiB/s, done.
Resolving deltas: 100% (5/5), done.
ubuntu@ip-172-31-37-27:~$ cd aws
ubuntu@ip-172-31-37-27:~/aws$ ls
index.js  package.json
ubuntu@ip-172-31-37-27:~/aws$
```

PHASE 5 – Configure Security Group

Step 1: Open EC2 Dashboard

1. Login to AWS Console
2. Go to EC2
3. Click Instance
4. Click on Security tab

The screenshot shows the AWS Management Console with the EC2 dashboard. The left sidebar shows the navigation menu with 'Instances' selected. The main content area shows the 'Security' tab for an instance. The 'Security details' section shows the IAM role, Owner ID, and Launch time. The 'Inbound rules' section shows a table of rules.

Name	Security group rule ID	Port range	Protocol	Source
-	sgr-074e3c77083dd8d3b	443	TCP	0.0.0.0/0
-	sgr-049dcb4a852ae0f8e	22	TCP	0.0.0.0/0
-	sgr-0ffc55a64c3479271	80	TCP	0.0.0.0/0

Step 2: Open Security Group

5. Under Security Groups, click on the linked Security Group name [example: **sg-0b2aeedb669246ce8 (launch-wizard-4)**] .
6. This opens the Security Group configuration page.

Step 3: Edit Inbound Rules

7. Click Edit inbound rules

The screenshot shows the AWS Management Console with the Security Groups dashboard. The left sidebar shows the navigation menu with 'Security Groups' selected. The main content area shows the configuration page for the security group. The 'Details' section shows the Security group name, ID, Description, VPC ID, Owner, Inbound rules count, and Outbound rules count. The 'Inbound rules' section shows a table of rules.

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-074e3c77083dd8d3b	IPv4	HTTPS	TCP	443
-	sgr-049dcb4a852ae0f8e	IPv4	SSH	TCP	22
-	sgr-0ffc55a64c3479271	IPv4	HTTP	TCP	80

8. Click Add Rule

Add the following rules in image:

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EC2 > Security Groups > sg-0b2aeedb669246ce8 - launch-wizard-4 > Edit inbound rules

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID	Type	Protocol	Port range	Source	Description - optional	
sgr-074e3c77083dd8d3b	HTTPS	TCP	443	Cust...	0.0.0.0/0	Delete
sgr-049dcb4a852ae0fbe	SSH	TCP	22	Cust...	0.0.0.0/0	Delete
sgr-0ffc55a64c3479271	HTTP	TCP	80	Cust...	0.0.0.0/0	Delete
-	Custom TCP	TCP	4000	Any...	0.0.0.0/0	Delete

[Add rule](#)

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Purpose: Allows Node.js application running on port 4000 to be accessible.

Step 4: Save Rules

Click Save Rules.

Now the instance allows:

1. SSH connection
2. TTP access
3. HTTPS access
4. Node app port (4000)

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EC2 > Security Groups > sg-0b2aeedb669246ce8 - launch-wizard-4

EC2

- Dashboard
- AWS Global View
- Events
- Instances**
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - Capacity Manager
- Images**
 - AMIs
 - AMI Catalog
- Elastic Block Store**
 - Volumes
 - Snapshots

Inbound security group rules successfully modified on security group (sg-0b2aeedb669246ce8 | launch-wizard-4)

sg-0b2aeedb669246ce8 - launch-wizard-4

[Details](#)

Security group name launch-wizard-4	Security group ID sg-0b2aeedb669246ce8	Description launch-wizard-4 created 2026-02-18T15:43:13.239Z	VPC ID vpc-035f96df3c676ba1d
Owner 780886633843	Inbound rules count 4 Permission entries	Outbound rules count 1 Permission entry	

[Inbound rules](#) | [Outbound rules](#) | [Sharing](#) | [VPC associations](#) | [Related resources - new](#) | [Tags](#)

Inbound rules (4)

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-074e3c77083dd8d3b	IPv4	HTTPS	TCP	443
-	sgr-0329c9d4029e2fff9	IPv4	Custom TCP	TCP	4000
-	sgr-049dcb4a852ae0fbe	IPv4	SSH	TCP	22

PHASE 6 – Run the Application

Step 1: Start Application

npm install

then,

node index.js

OR

npm start

If successful, terminal shows:

```
ubuntu@ip-172-31-37-27:~/aws$ node index.js
Started server
```

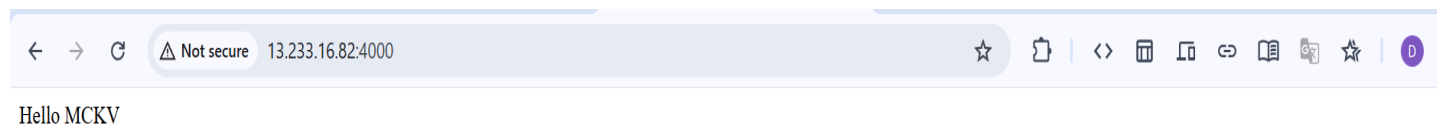
The static website will load successfully.

Step 2: Access Application in Browser

Open a browser and enter:

http://YOUR_PUBLIC_IP:4000

➤ In My Case: <http://13.233.16.82:4000>



PHASE 7 – Stop and Terminate Instance

Step 1: Stop the EC2 Instance

1. Go back to the EC2 Instances page.
2. Select the instance.
3. Click Instance state → Stop instance.
4. Confirm by clicking Stop.

A screenshot of the AWS Management Console. The top navigation bar shows the AWS logo, a search bar, and the user's name 'Diptadeep Ghosh'. The left sidebar shows the 'EC2' menu with options like 'Dashboard', 'AWS Global View', 'Events', 'Instances', 'Instance Types', 'Launch Templates', 'Spot Requests', and 'Savings Plans'. The main content area shows the 'Instances (1/1)' page. A table lists the instance 'WebServer-AS...' with ID 'i-00ee2b4655a95b9a5' and state 'Running'. A dropdown menu is open under the 'Instance state' column, showing options: 'Stop instance', 'Start instance', 'Reboot instance', 'Hibernate instance', and 'Terminate (delete) instance'. Below the table, a 'Stop instance' modal is displayed. It explains that stopping the instance allows for cost reduction and troubleshooting. It shows the instance ID 'i-00ee2b4655a95b9a5 (WebServer-ASSG9)' with 'Stop protection' set to 'Disabled' and a 'Result' of 'Can stop'. It also lists 'Associated resources' and a warning that the user will be billed for associated Elastic IP addresses and EBS volumes. At the bottom, there is a 'Skip OS shutdown' option and buttons for 'Cancel' and 'Stop'.

Step 2: Terminate the EC2 Instance

1. Select the same instance again.
2. Click Instance state → Terminate (delete) instance.

The screenshot shows the AWS Management Console interface for EC2 Instances. A green notification bar at the top indicates 'Successfully initiated stopping of i-00ee2b4655a95b9a5'. The 'Instances (1/1)' table shows one instance, 'WebServer-AS...', with ID 'i-00ee2b4655a95b9a5' and state 'Stopping'. The 'Instance state' dropdown menu is open, showing options: 'Force stop instance', 'Start instance', 'Reboot instance', 'Hibernate instance', and 'Terminate (delete) instance'.

The 'Terminate (delete) instance' dialog is shown. It includes a warning: 'On an EBS-backed instance, the default action is for the root EBS volume to be deleted when the instance is terminated. Storage on any local drives will be lost.' Below this, it asks 'Are you sure you want to terminate these instances?' and shows the instance ID 'i-00ee2b4655a95b9a5 (WebServer-ASSG9)' with 'Termination protection' set to 'Disabled'. A note states: 'To confirm that you want to delete the instances, choose the terminate button below. Instances with termination protection enabled will not be terminated. Terminating the instance cannot be undone.' There is a checkbox for 'Skip OS shutdown' with the text: 'This option skips the graceful OS shutdown process. Use only when your instance must be stopped immediately, such as during an emergency or failover.' At the bottom, there are 'Cancel' and 'Terminate (delete)' buttons.

The screenshot shows the AWS Management Console interface for EC2 Instances after the termination process. A green notification bar at the top states 'Successfully initiated termination (deletion) of i-00ee2b4655a95b9a5'. Below it, another notification bar states 'Successfully initiated stopping of i-00ee2b4655a95b9a5'. The 'Instances (1/1)' table shows the instance 'WebServer-AS...' with ID 'i-00ee2b4655a95b9a5' and state 'Terminated'. The 'Status check' column shows '3/3 checks passed'.

Conclusion

This assignment demonstrated practical understanding of:

- AWS EC2 instance management
- Linux server configuration
- Node.js deployment
- Security group configuration
- Reverse proxy setup
- Production process management